## A. Complete Letter-Level Results

Table 3 includes every letter, not only selected gains or regressions. Metrics are percentages; $\Delta n$ is the change in correct true-letter images. "Base" is the project-profile SVM with 820 Shin inputs; "Angles" adds the 15 world-joint measurements. These are the earlier +67 results, not the same-coordinate +4 control in Appendix K. Support varies because detection filtering is not uniform across letters. Precision measures the correctness of predictions of a letter, while recall measures recovery of its true images.

*Table 3.* All-letter pooled OOF comparison.

| | | Precision (%) | | Recall (%) | | F1 (%) | | |
|---|---|---|---|---|---|---|---|---|
| Letter | Images | Base | Angles | Base | Angles | Base | Angles | $\Delta n$ |
| A | 2,188 | 98.32 | 98.00 | 98.67 | 98.72 | 98.49 | 98.36 | +1 |
| B | 2,204 | 99.28 | 99.14 | 99.77 | 99.73 | 99.52 | 99.43 | -1 |
| C | 1,912 | 99.17 | 99.07 | 99.84 | 99.84 | 99.50 | 99.45 | +0 |
| D | 2,456 | 99.01 | 98.81 | 97.68 | 97.68 | 98.34 | 98.24 | +0 |
| E | 2,308 | 98.24 | 98.58 | 99.00 | 99.22 | 98.62 | 98.90 | +5 |
| F | 2,859 | 99.23 | 99.26 | 98.60 | 98.57 | 98.91 | 98.91 | -1 |
| G | 2,430 | 98.89 | 98.94 | 99.42 | 99.42 | 99.16 | 99.18 | +0 |
| H | 2,371 | 98.81 | 98.64 | 98.14 | 98.19 | 98.48 | 98.41 | +1 |
| I | 2,380 | 97.29 | 97.46 | 96.68 | 96.76 | 96.99 | 97.11 | +2 |
| J | 2,576 | 98.07 | 98.19 | 96.78 | 96.78 | 97.42 | 97.48 | +0 |
| K | 2,701 | 98.92 | 99.18 | 98.26 | 98.37 | 98.59 | 98.77 | +3 |
| L | 2,516 | 99.72 | 99.80 | 99.64 | 99.60 | 99.68 | 99.70 | -1 |
| M | 1,602 | 88.74 | 89.11 | 90.01 | 91.95 | 89.37 | 90.51 | +31 |
| N | 1,280 | 88.89 | 91.32 | 88.12 | 87.97 | 88.51 | 89.61 | -2 |
| O | 2,264 | 96.86 | 96.73 | 98.19 | 97.88 | 97.52 | 97.30 | -7 |
| P | 2,041 | 97.96 | 97.82 | 98.63 | 98.78 | 98.29 | 98.29 | +3 |
| Q | 2,106 | 98.75 | 98.94 | 97.72 | 97.72 | 98.23 | 98.33 | +0 |
| R | 2,543 | 96.23 | 96.87 | 93.43 | 93.83 | 94.81 | 95.33 | +10 |
| S | 2,559 | 96.87 | 97.15 | 98.09 | 98.40 | 97.48 | 97.77 | +8 |
| T | 2,354 | 99.14 | 99.06 | 98.43 | 98.47 | 98.78 | 98.76 | +1 |
| U | 2,513 | 92.05 | 92.11 | 96.70 | 96.58 | 94.31 | 94.29 | -3 |
| V | 2,552 | 98.20 | 98.32 | 98.24 | 98.43 | 98.22 | 98.37 | +5 |
| W | 2,456 | 98.41 | 98.21 | 98.25 | 98.33 | 98.33 | 98.27 | +2 |
| X | 2,157 | 96.50 | 96.86 | 97.03 | 97.22 | 96.76 | 97.04 | +4 |
| Y | 2,580 | 99.46 | 99.50 | 99.46 | 99.38 | 99.46 | 99.44 | -2 |
| Z | 2,343 | 99.09 | 98.96 | 97.44 | 97.78 | 98.26 | 98.37 | +8 |

The remaining angle-model errors include R→U (142), N→M (141), M→N (102), U→R (44), D→O (41), Q→P (41) and O→D (19). These counts suggest several candidate investigations; prioritizing only D/O would ignore larger residual error sources. Full ordered error pairs are retained in `results/angles/confusions/all_error_pairs.csv`.

## B. Full Confusion Matrices

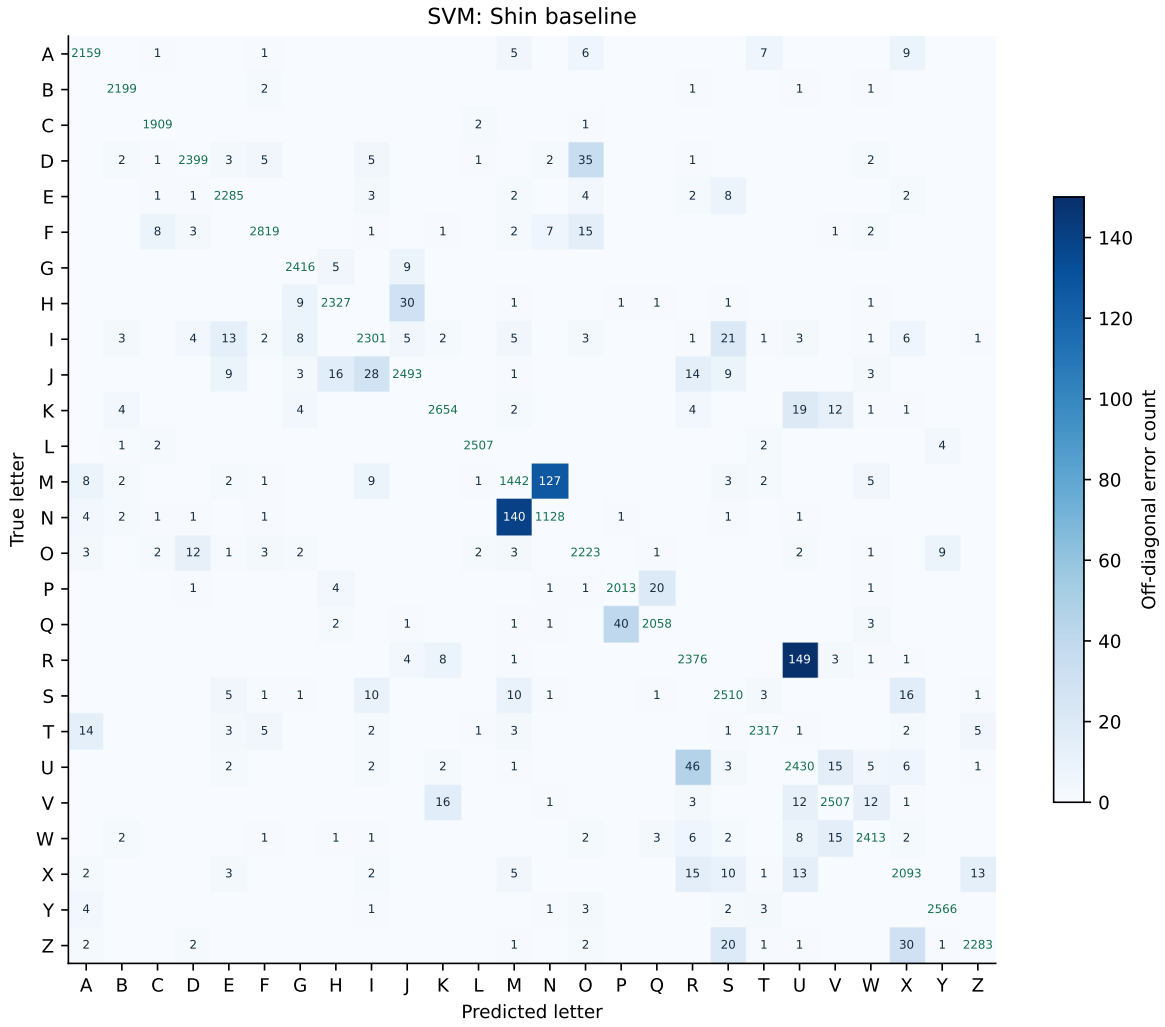



*Figure 3.* SVM baseline OOF confusion counts for all 26 letters. Diagonal counts are green; shading is restricted to off-diagonal errors to keep smaller confusions visible. Rows are true labels and columns predictions.

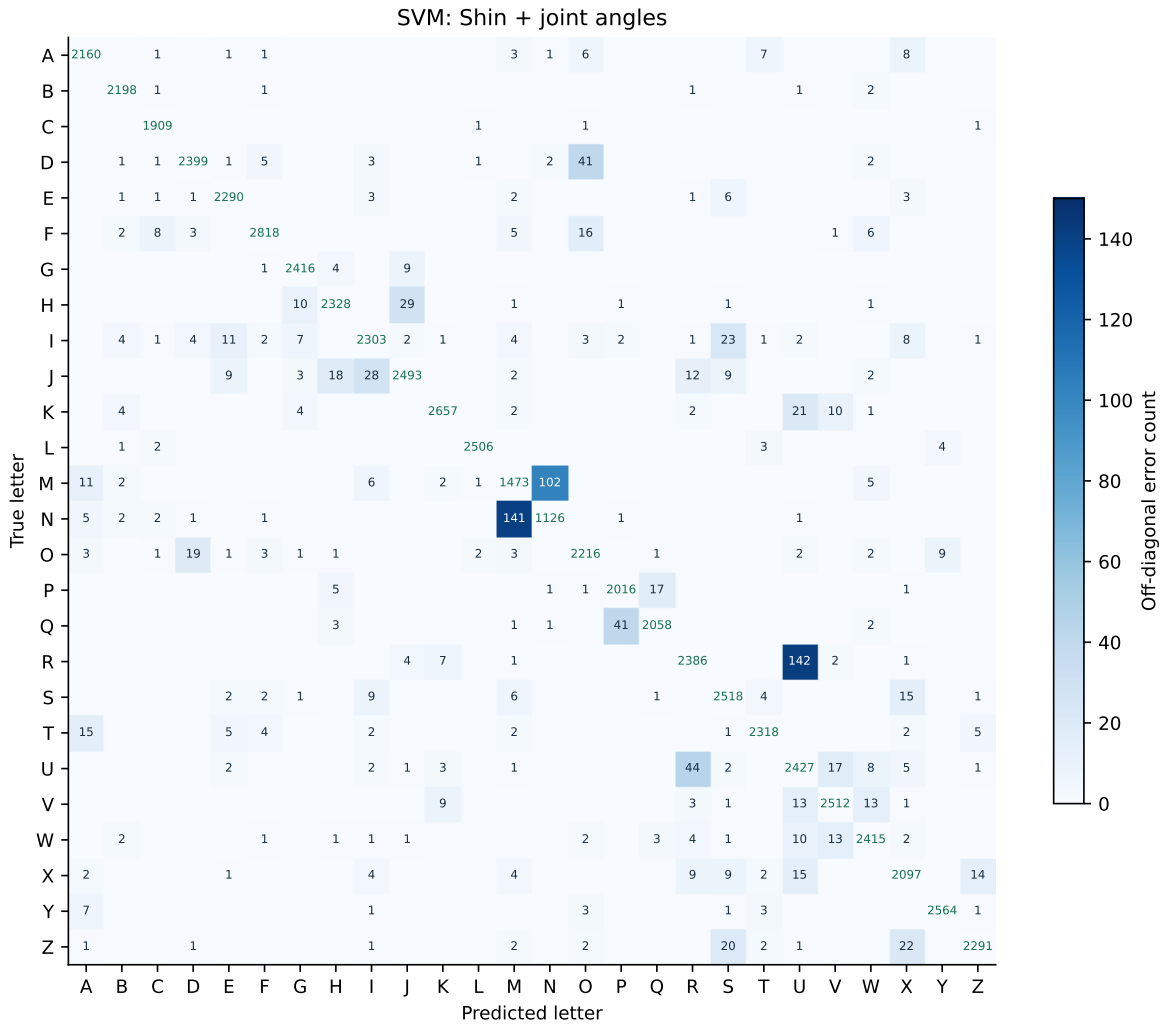



*Figure 4.* Same images and folds with continuous angles added. Compare, for example, the M/N, R/U and D/O cells with the baseline matrix. Counts, rather than rounded percentages alone, expose small regressions.

## C. Qualitative Examples and Feature Definitions

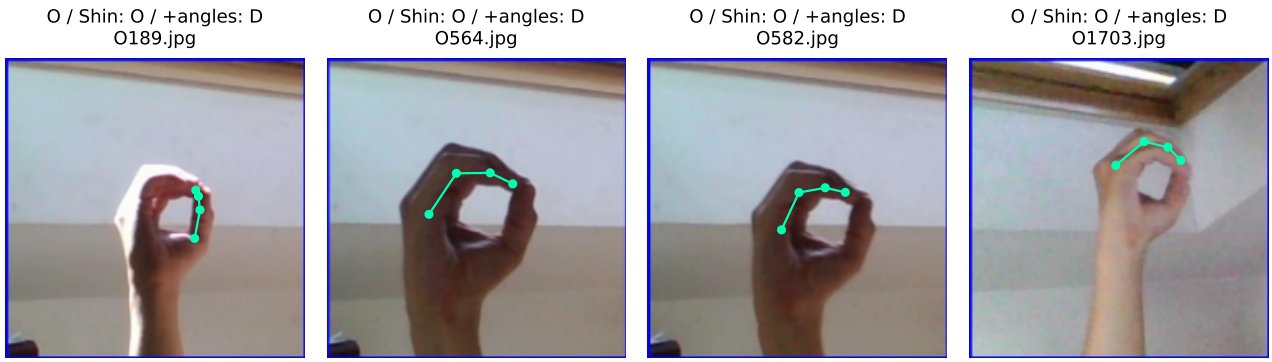



*Figure 5.* Four actual Kaggle O images newly classified as D after adding angles; the index-finger landmarks are overlaid. These are static dataset errors, not recorded webcam rotation tests. Dataset labels are retained; no claim of manually revalidated signing correctness is made.

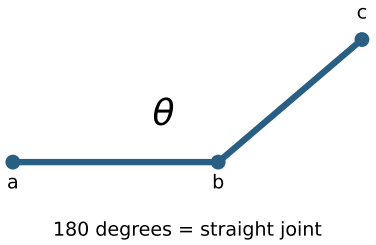



*Figure 6.* Interior-angle convention for a joint centred at $b$. A straight joint approaches $180^\circ$; flexion reduces the angle.

*Table 4.* Landmark triplets for the 14 interior angles, using MediaPipe indices.

| Finger | MCP | PIP | DIP / IP |
|---|---|---|---|
| Thumb | (1,2,3) | — | IP: (2,3,4) |
| Index | (0,5,6) | (5,6,7) | (6,7,8) |
| Middle | (0,9,10) | (9,10,11) | (10,11,12) |
| Ring | (0,13,14) | (13,14,15) | (14,15,16) |
| Little | (0,17,18) | (17,18,19) | (18,19,20) |

The fifteenth input is signed thumb-MCP deviation from the palm-oriented geometry implementation. The complete measured feature schema, unknown-state handling and thresholds are stored in `research/schema.py`, `research/geometry.py` and `config/thresholds.json`. The workbook parser preserves slash-separated alternatives as allowed sets. The user's workbook is unchanged; `config/proposal_changes.json` contains the 49 proposed relaxations. They are applied in memory for the historical compatibility-score configuration, without retaining a second spreadsheet.

For U/V/K-style spread, fingers must be sufficiently separated and approximately coplanar; they need not be strictly parallel. Crossing detection uses a palm-oriented swap/depth proxy rather than assuming any two projected image lines prove a crossing. These engineering rules are auditable hypotheses and are not validated ground-truth annotations of anatomy.

## D. Paired Statistics and Weighting Details

Relative to the baseline, adding world-joint measurements fixes 143 formerly incorrect images and loses 76 formerly correct images, a net gain of 67. The descriptive accuracy change is 0.1112 percentage points. Six filename blocks give the changes below. Resampling these blocks with replacement 10,000 times (seed 42) gives a central 95% sensitivity range of 0.0374–0.1947 percentage points. This is a descriptive stability check, not a valid independent-test confidence interval: there are only six non-person groups, possible cross-block correlation, overlapping training sets and exploratory model selection. We do not report a confirmatory $p$-value. The primary weighting study evaluated 0.25/0.5/1/2 multipliers for

*Table 5.* Paired world-joint-model changes by filename block.

| Block | Images | Net correct | Change (pp) |
|---|---|---|---|
| 0000 | 10,504 | +4 | +0.0381 |
| 0001 | 11,399 | +27 | +0.2369 |
| 0002 | 12,101 | +2 | +0.0165 |
| 0003 | 12,255 | +5 | +0.0408 |
| 0004 | 10,009 | +23 | +0.2298 |
| 0005 | 3,983 | +6 | +0.1506 |

continuous angles, categorical encodings and compatibility scores, with a single coordinate pass. In the targeted study, nested PIP weights were 2, 2, 0.5, 0.5 and 0.5 over the five outer folds. The selected MCP/PIP/DIP triples were (0.5,2,1), (1,2,0.5), (0.5,0.5,1), (2,0.5,1) and (1,0.5,1). These varying choices further caution against a universal rule that the index finger should receive higher weight.

*Table 6.* Targeted SVM weighting on the 835-input angle configuration. Errors are directed true-label to prediction counts.

| Configuration | Accuracy (%) | Macro-F1 | O→D | N→M |
|---|---|---|---|---|
| Unweighted | 97.74776 | .975244 | 19 | 141 |
| PIP weight 2 | 97.74941 | .975241 | 23 | 143 |
| Nested PIP | 97.75273 | .975265 | 16 | 143 |
| Nested MCP/PIP/DIP | 97.73946 | .975073 | 15 | 146 |

## E. Verification and Reproducibility

The source repository retains one training dataset, the user's workbook, fixed thresholds and proposal changes, the compact standalone implementation, necessary study/live tools and tests, and compact measured evidence. Bulky model checkpoints, superseded datasets and obsolete interfaces are moved to a separate reversible archive; none is needed to interpret the reported tables. A central cached input matrix and sample/group metadata live inside the retained dataset directory. `REPRODUCE.md` documents setup, inventory validation, rebuilding descriptors from frozen landmarks, model comparisons, permutation analysis, ablations, tuning and report compilation.

The verification checks reconcile stored predictions with pooled accuracy, macro-F1, per-letter metrics and confusion totals; check one OOF prediction per image per model/variant; verify shared fold assignments and disjoint outer groups; and test feature widths, angle conventions, slash parsing, train-only preprocessing and index-weight application. Reconstructed descriptors are compared with the historical numeric cache. These checks support computational consistency; they do not validate source labels, estimated 3D anatomy or generalization to unseen signers. Large research experiments were completed before cleanup; refactoring preserves their numerical inputs and fixed profiles rather than silently reporting newly selected runs.

## F. Rubric Coverage and Outstanding Research

| Rubric component | Coverage and limits |
|---|---|
| Abstract | Motivation, objective, methods, numerical outcomes and scope. |
| Introduction | Field, motivation, contribution, questions and structure. |
| Related work | Baseline geometry, landmark estimation, alternative sensing/classification, permutation reliance and selection bias. |
| Methods | Representations, angle equation, model objectives, weighting and preprocessing. |
| Experiments/analysis | Matched baseline, dataset filtering, ablations, two algorithms, nested weighting, per-letter errors and descriptive paired analysis. No untouched final test or confirmatory significance claim. |
| Discussion | Meaning, strengths, limitations, ethics and AI-generation disclosure. |
| Conclusions | No consistent benefit from interpretable additions; distinguish the small world-coordinate gain from the near-zero same-coordinate result. |

The main report now answers the explicit-feature comparison question through RQ1–RQ3. Later webcam experiments are supporting diagnostics, not unfinished conditions for that answer. A prospective signer-separated evaluation and independent review of the workbook remain valuable follow-ups. The detailed rubric assessment is retained separately in `paper/RUBRIC_REVIEW.md`.

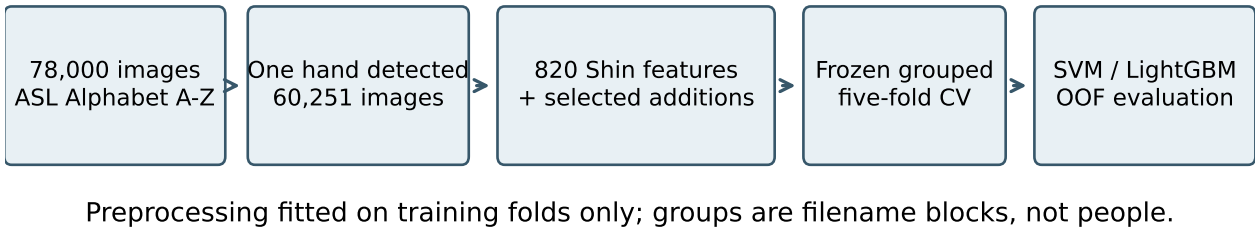



*Figure 7.* Evaluation pipeline. Every compared model uses the same 60,251 accepted images and frozen outer folds. Detection coverage and conditional classification accuracy are separate quantities.

## G. Technical Method Details

### G.1. Landmarks and Baseline

Images are processed with MediaPipe Hand Landmarker. The intended extraction settings are IMAGE mode, up to two hands, and detection/presence thresholds of 0.5; only exactly one accepted hand is used. Historical records contain the MediaPipe version and landmarks, but not the threshold values. Thus default-setting provenance cannot be independently confirmed for the historical cache. We preserve these landmarks for all comparisons rather than rerun detection differently per model.

The Shin implementation follows the paper's pair list: 190 pairwise distance features and 630 axis-direction features, for 820 features overall. Distance features are normalized per image; direction features are cosine-based encodings of inter-landmark vectors relative to axes. The local implementation preserves the paper's listed pair exclusions. The original additions use estimated world landmarks, whereas the baseline uses image landmarks. Appendix K adds a matched image-coordinate control. Shin directions already contain information from which interior joint angles can be derived; explicit measurements test representation rather than new sensing information.

### G.2. Interpretable Additions

For consecutive landmarks $a, b, c$, an interior bending angle is

$$\theta(a,b,c) = \arccos \frac{(a-b)^\top (c-b)}{\|a-b\|\,\|c-b\|} \frac{180}{\pi}. \tag{1}$$

A straight joint approaches $180^\circ$. We use thumb MCP/IP and MCP/PIP/DIP for the other four fingers: 14 interior angles. A signed thumb-MCP deviation adds a fifteenth measurement. MCP measurements are landmark-based proxies, not calibrated clinical joint measurements. "Angles" in the original comparison means all 15 world-derived additions. Appendix K labels the same-coordinate measurements separately.

Categorical features encode bent, partly bent and extended joints; a binary B-shaped/non-B-shaped thumb MCP; and spread, suspected crossing and thumb contact relations. An unknown state is retained. Thresholds are fixed across folds. Non-thumb flexion uses $\leq 130^\circ$ for bent and $\geq 150^\circ$ for extended, with intermediate values partly bent. Finger spread requires separation in approximately the same plane; splaying is allowed. Crossing and contact are geometric proxies and remain sensitive to occlusion and depth estimates.

The workbook gives the allowed categories for each letter. A slash means a set of acceptable alternatives, not a new combined class. One-hot measured categories yield 70 inputs. For each candidate letter $\ell$, a compatibility score averages the proportion of specified workbook cells whose allowed set contains the measured category. This produces 26 scores. All 26 are computed for every image without accessing its true label. A fixed 49-cell slash-relaxation proposal was developed in earlier exploratory work; its changes are preserved separately from the user's original workbook. This reference design was not nested within the later evaluation.

### G.3. Models, Scaling and Weighting

The primary SVM uses an RBF kernel, $C = 10$, `gamma=scale` and balanced class weights. This differs from the paper-profile $C = 1000$ with no class balancing. Its soft-margin objective balances margin size against weighted classification slack. For input blocks $z_b$ and weights $w_b$, the RBF similarity is

$$K(x, x') = \exp\left(-\gamma \sum_b w_b^2 \|z_b(x) - z_b(x')\|^2\right). \tag{2}$$

For each binary letter pair, the kernel SVM minimizes the regularized hinge loss

$$\frac{1}{2}\|w\|^2 + C \sum_i \omega_i \max(0, 1 - y_i(w^\top \phi(x_i) + b)),$$

where $y_i \in \{-1, 1\}$, $\phi$ is the kernel feature mapping and $\omega_i$ combines class and sample weights. The multiclass decision uses pairwise voting. LightGBM instead optimizes multiclass log loss with tree-complexity control. These objectives were not modified to prefer individual letters.

Thus a multiplier of two contributes a factor of four inside the squared distance, while `gamma=scale` also changes with the input variance. Angle median imputation and standardization are fitted within each training fold; baseline descriptors and categorical/compatibility values pass through unchanged. No missing angles occurred in the retained cache.

LightGBM optimizes multiclass log loss through boosted trees, using 100 estimators, 31 leaves, learning rate 0.1 and minimum child size 20. Its fixed project profile is not a replication of the paper's tuned LightGBM settings. Group weighting searches use inner-fold macro-F1, breaking ties by accuracy. They make one coordinate pass, not an exhaustive global search.

## H. Earlier Experimental Settings and Results

### H.1. Implementation and Experimental Sequence

Experiments use Python 3.12, MediaPipe 0.10.35, scikit-learn 1.8.0 and LightGBM 4.7.0. An earlier three-fold pilot tested 81 continuous/binary finger-geometry descriptors, alone and combined with baseline descriptors. This differs from the later 70-column categorical encoding and is not mixed into the five-fold table. The pilot feature-only model reached 93.71% mean fold accuracy, versus 97.43% for the baseline; combining them did not improve the baseline. It did not isolate replacement of only the 190 distances while retaining all 630 directions.

The main sequence compares the 820 baseline, all 111 additions (931 total), then isolated angle and compatibility-score additions. Grouped permutation analysis shuffles whole related columns jointly on held-out data, five repeats with at most 100 images per letter per fold. Ablations remove a block and refit. These analyses share the fixed full-input model and should not be interpreted as interchangeable measures.

Block weighting tries 0.25, 0.5, 1 and 2. Targeted index MCP/PIP/DIP weighting tries 0.5, 1 and 2. Both use five outer and three inner group folds, with inner training and validation capped at 150 images per letter for cost. The final outer fits use all outer-training images. This is bounded exploratory tuning, not exhaustive hyperparameter optimization.

### H.2. Isolated SVM and LightGBM results

*Table 7.* Detailed original isolated comparison; these results concern matched accepted images.

| Representation | Inputs | SVM acc. (%) | SVM macro-F1 | LGBM acc. (%) | LGBM macro-F1 |
|---|---|---|---|---|---|
| Shin | 820 | 97.637 | 0.97386 | 97.097 | 0.96784 |
| Shin + angles | 835 | 97.748 | 0.97524 | 97.017 | 0.96692 |
| Shin + Excel | 846 | 97.648 | 0.97400 | 97.119 | 0.96815 |
| Shin + angles + Excel | 861 | 97.744 | 0.97515 | 97.129 | 0.96824 |

The full 931-input SVM scored 97.62826% and LightGBM 97.16353%. Removing categorical one-hot columns produced 97.74444% for SVM, matching the isolated angle-plus-workbook model. In the grouped permutation analysis, shuffling

angles reduced SVM macro-F1 by 0.570 percentage points, compared with 0.049 for categories and 0.008 for workbook compatibility scores. For LightGBM these drops were 0.031 and 0.200 for angles and compatibility scores. These overlapping reliance measures are not additive or causal.

The source article reports 87.60% ASL Alphabet combined-feature SVM accuracy, no matching F1 metric, and a held-out evaluation with tuning rather than this filtered grouped protocol. It also states 780,000 images inconsistently with 26 times 3,000; the actual inventory is 78,000. The local paper-profile SVM ($C = 1000$, no balancing) reached 97.66145%; changing this setting alone does not reconcile the protocols. Detection, data selection and split differences have not been isolated experimentally.

## I. Tilt Augmentation and Recorded Development Study

The original training input contains 820 Shin values plus 15 angles. For each of the 210 unit direction vectors $d_{ij}$, the augmentation uses

$$R_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha & -\sin\alpha \\ 0 & \sin\alpha & \cos\alpha \end{bmatrix}, \qquad d'_{ij} = R_x(\alpha) d_{ij}, \qquad \alpha \in \{-15^\circ, 0, 15^\circ\}.$$

Only input columns 190–819 change (zero-based indexing). The first 190 normalized distances and final 15 intrinsic measurements stay fixed. This rotation is performed in normalized landmark-feature coordinates and is an approximation, not a full physical wrist or perspective model. It does not simulate occlusions or detector errors.

The all-letter run uses 180,753 fitting rows from 60,251 originals, with weight $1/3$ per row. Class balancing and gamma=scale are applied through the fixed SVM pipeline. The C-only run adds two copies of each of the 1,912 C originals, producing 64,075 fitting rows; 58,339 other images remain single examples. Each C source retains total sample weight one, and other rows have weight one. For selective augmentation, preprocessing, balanced class weights and the resolved gamma value are fitted on original fitting rows, avoiding unintended class-frequency or scaling changes. The C-only full-data resolved gamma and class weights match the baseline exactly. The earlier all-letter run computed gamma over its augmented matrix, so the three-way comparison is not a strict single-factor ablation of that numeric parameter.

Both variants were tested on the same original five grouped folds. Augmentation occurs only after splitting, inside the fitting partition. Each variant yields 60,251 original validation predictions. The all-letter variant recovers 57 errors and loses 84 correct predictions: 58,867 correct versus 58,894 for the angle baseline. Its accuracy is 97.70294269% and macro-F1 0.974685870. The C-only variant has the same 58,894 correct images as baseline, with zero recoveries or losses and three wrong-to-wrong label changes; macro-F1 changes from 0.975243912 to 0.975244649. This difference is practically negligible.

The guided development session contains 27 planned captures: C/D/O, three qualitative wrist positions and three repetitions. The intended label and step are fixed; predictions are hidden during collection. All 27 have accepted detections. The session predates candidate fitting and never enters training. The reference Shin-only full-data fit scores 17/27 and C/D/O macro-F1 0.609524; the joint-angle fit scores 17/27 and 0.610035. All-letter tilt scores 18/27 and 0.642439; C-only tilt returns the same predictions as the joint-angle baseline. Joint-angle-model counts by intended letter are C=3/9, D=8/9, O=6/9; all-letter tilt changes only the number correct for O to 7/9. One wrong C prediction also changes from D to O, lowering O precision while improving its recall.

These are development outcomes, not independent test results. The reserved evaluation directory contained zero completed captures when these experiments were run. The previously inspected second-person O example is an exploratory illustration and is not added to the primary 27. No angle/axis search was conducted within either augmentation run, but the research sequence as a whole is exploratory.

The main input/model/image hashes and saved predictions were audited. In the 28 records comprising the development round and extra O example, all image hashes matched and all 835 feature vectors and predictions were reconstructed. The additional reference model is fitted on the same 60,251 original images using only 820 inputs; its live-bundle schema is distinguished from the 835-input models. Snapshot exports retain the selected model's hash, input variant and matching feature width.

Reproduction commands, evidence paths and the live model-label mapping are documented in `REPRODUCE.md`.

## J. Later Webcam and Representation Diagnostics

These experiments followed the primary feature comparison. They are retained for transparency and hypothesis generation; they do not expand the main report's research question into a claim that webcam or viewpoint problems have been solved. Stored development errors were reused to motivate successive changes, so their outcomes are not independent final-test evidence.

### J.1. Larger G/H Rotations and Projected Angles

A selective G/H augmentation rotates only the 630 image-axis direction values around the camera-$y$ axis by $-90°$, $90°$ and $180°$, retaining each original. There are 4,801 G/H originals and 74,654 fitting rows overall. Each selected source contributes four rows of weight 0.25; other classes occur once with weight one. Preprocessing, balanced class weights and resolved gamma are determined on the original fitting partition. This is a numerical landmark-feature rotation, not a realistic palm/back-of-hand rendering or occlusion simulation.

A separate unaugmented candidate adds 14 internal angles measured from projected image landmarks, after converting normalized $x, y$ into pixels with the detector image width and height. It has 849 inputs: the original 820, 15 estimated 3D measurements and 14 projected angles. The same function is used for training and live inference. Median imputation and standard scaling of all 29 angle inputs are fitted only on each fitting partition. Camera captures are not used for fitting either candidate.

*Table 8.* Supporting 26-class experiments using the same frozen five folds. These extend the research sequence beyond the main feature comparison.

| Model | Correct / 60,251 | Accuracy (%) | Macro-F1 |
|---|---|---|---|
| Shin + 3D measurements | 58,894 | 97.74776 | .97524391 |
| Selective G/H rotation | 58,888 | 97.73780 | .97516874 |
| Shin + 3D + projected angles | 58,861 | 97.69298 | .97465145 |

Selective G/H rotation fixes nine formerly wrong predictions and loses 15 correct ones. The projected-angle model fixes 70 and loses 103. Its five-fold accuracy declines in four folds and improves in one. Neither is an overall improvement on the 835-input reference. On the 27 earlier C/D/O development captures, the projected-angle model returns 16 correct versus 17 for the angle baseline.

The selected GH-handrug development round contains 11 errors from the G/H-rotation model: G→H, C→D, F→B, K→H, N→M, O→D, P→H, S→E, T→L, X→H and Y→A. Replaying the original 835-input model on these same landmarks recovers K and X; adding projected angles also recovers T. The newest G remains misclassified. This is a selected-error replay, not a representative live-accuracy comparison.

### J.2. G/H-Only Diagnostic Classifiers

The primary 26-class angle SVM recognizes 2,416/2,430 accepted G images correctly (99.424% recall). Its 14 true-G errors are nine J, four H and one F. For H, 2,328/2,371 are correct (98.186%); ten are G and 29 J, with four other errors. Thus strong accepted-image dataset recognition coexists with the recorded webcam failures.

Three fixed binary SVMs were fitted using only 3D angles, only projected angles or both. The same frozen folds were restricted to their 4,801 G/H images. Every variant used training-only median imputation and standard scaling, with RBF $C = 10$, balanced classes and gamma=scale. They are easier two-label tasks with different inputs, not replacements for the 26-class recognizer.

All three fuller binary variants still classify the newest selected G as H. Its middle-PIP estimate is $95.3°$, near the training-G median $86.7°$ and below the H median $166.2°$. Other measurements disagree with typical G: the index PIP is $134.9°$ against a G median $172.6°$, while middle DIP is $161.0°$ against medians $113.2°$ for G and $170.4°$ for H. Its signed thumb deviation is $-1.5°$ against medians around 36–37° for both. The classifier sees a combination, not just a count of straight fingers.

The two-input middle-PIP ablation recovers the two selected G examples and one H example, but has lower grouped

*Table 9.* Binary G/H diagnostics. The fourth row is an explicitly post-hoc, targeted ablation after inspecting the first three.

| Inputs | Number | Accuracy (%) | Macro-F1 |
|---|---|---|---|
| 3D measurements | 15 | 96.563 | .96563 |
| Projected 2D angles | 14 | 95.793 | .95791 |
| Both representations | 29 | 96.584 | .96584 |
| Middle PIP only, 3D + 2D | 2 | 94.085 | .94084 |

dataset performance. It was proposed after inspecting these known failures and is exploratory. A later selected H with visible palm is called G by all six 26-class variants, although its estimated index/middle PIP angles are $165.5°/159.8°$. The binary diagnostics call it H. This counterexample shows that palm versus back-of-hand viewpoint alone cannot explain every recorded error.

A four-view protocol for three participants (G/H, palm, oblique palm, oblique back-of-hand, back-of-hand; five repetitions) has been implemented, but no completed cohort evaluation is claimed here. The qualitative views are not calibrated physical angles. New balanced and signer-separated observations would be needed for a prospective viewpoint comparison.

Evidence and reproduction commands are in `results/rotation_GH_01/`, `results/angles_2d_01/`, `results/gh_classifier_01/`, `results/gh_middle_01/` and `REPRODUCE.md`. Saved image/model hashes, raw measurements and predictions support computational replay, not anatomical or linguistic ground truth.

## K. Direction Features and Explicit Joint Measurements

### K.1. Purpose and Relationship to Shin

The original motivation was to test descriptions such as bent, extended, spread and touching. Continuous measurements are a comparison with these categories. Shin's 630 direction cosines already encode all 210 landmark-pair directions. For $u = (a - b)/\|a - b\|$ and $v = (c - b)/\|c - b\|$, the joint angle is $\arccos(u^\top v)$: it is obtainable from those descriptors with appropriate pair signs. Thresholding an angle into a category likewise supplies no independent sensor information.

Making a nonlinear relation explicit may help a classifier. The original additions also use estimated world landmarks, while Shin uses normalized image landmarks. The control separates coordinate source from joint representation; it does not assume that Shin omitted finger-bending information.

### K.2. Matched Design

The same 60,251 accepted images and five frozen filename-block folds are retained, without new detection, augmentation or tuning. The 15 measurements (14 interior angles plus signed thumb-MCP deviation) use Shin's exact normalized image $x, y, z$ coordinates without aspect correction. These are descriptors, not calibrated anatomical angles.

The reference has 190 distances and 630 directions (820 inputs). Replacement substitutes 15 joint measurements for directions (205 inputs); addition retains the reference (835). The earlier 835-input world-joint model provides context. Historical predictions are aligned by ID, label and fold, with source hashes checked. Imputation and standardization use training rows only. Fixed classifier settings match the original profiles; RBF `gamma=scale` adapts to each representation. These recipes are not equally optimized upper bounds.

*Table 10.* Coordinate control on matched held-out predictions. Fixed/lost/net refer to each learner's Shin reference. The world rows reproduce the original comparison; they are not new runs.

| Learner | Representation | Inputs | Acc. (%) | Macro-F1 | Fixed | Lost | Net |
|---|---|---|---|---|---|---|---|
| SVM | Reference | 820 | 97.63655 | 0.973862 | 0 | 0 | +0 |
| SVM | Distances + same-coordinate joints | 205 | 94.51129 | 0.943797 | 284 | 2167 | -1883 |
| SVM | Shin + same-coordinate joints | 835 | 97.64319 | 0.974028 | 89 | 85 | +4 |
| SVM | Shin + world joints | 835 | 97.74776 | 0.975244 | 143 | 76 | +67 |
| LightGBM | Reference | 820 | 97.09714 | 0.967843 | 0 | 0 | +0 |
| LightGBM | Distances + same-coordinate joints | 205 | 93.94201 | 0.937267 | 515 | 2416 | -1901 |
| LightGBM | Shin + same-coordinate joints | 835 | 97.04071 | 0.967403 | 138 | 172 | -34 |
| LightGBM | Shin + world joints | 835 | 97.01748 | 0.966920 | 98 | 146 | -48 |

### K.3. Interpretation and Evidence

Adding same-coordinate joints corrects 89 SVM errors and loses 85 correct predictions: net +4 (+0.0066 percentage points). Three folds improve and two worsen. The earlier world-joint result corrects 143 and loses 76: net +67 (+0.1112 points). Neither establishes new-person or webcam performance. LightGBM loses 34 correct predictions with same-coordinate joints and 48 with world joints.

Replacing directions loses a net 1,883 SVM and 1,901 LightGBM correct predictions, worsening every fold. For SVM, J-to-I errors increase from 28 to 531, I-to-J from 5 to 332, and H-to-U from zero to 230. Replacement removes axis-relative orientation as well as reducing feature count; this may discard useful letter distinctions. It is not evidence of faulty angle calculation, nor a controlled test of only rotation robustness.

The same-coordinate addition gives no convincing overall benefit under these recipes. The larger world-coordinate difference cannot be attributed to the bending formula alone; coordinate estimates, scaling and model geometry are possible contributors, not independently established causes. The full category/workbook package remains a separate comparison; this control does not isolate category-only predictive value.

The supplement `results/direction_joint_01/` contains per-letter metrics, confusion matrices, fold scores and per-image predictions. `research.paper_core_assets` generates paper rows after reconciling paired counts and historical scores. The implementation, runner and verifier are listed with commands in `REPRODUCE.md`. Computational checks do not establish independent scientific validation.

## L. Horizontal RGB Mirroring and Hand-Side Robustness

### L.1. Question, Motivation and Source Population

This follow-up asks whether adding horizontal RGB mirrors of training images changes precision and F1 on a controlled mixture of original and mirrored held-out images. It is separate from the primary comparison of interpretable feature representations. Left-handed fingerspelling is legitimate; Vicars (ASL University) describes its mirrored relationship with right-handed signing. Here the original static A–Z labels are retained under horizontal reflection. J and Z are static dataset poses, not tests of their movement.

The historical 78,000-image manifest contains 60,251 accepted single-hand detections: 57,424 are labelled Right by MediaPipe (95.31%) and 2,827 Left (4.69%). The remaining 17,749 have no accepted single-hand estimate. These are detector predictions about depicted hands, not verified annotations or evidence of the signer's preferred hand. The dataset is therefore predominantly detector-labelled Right among accepted images; the prevalence over all source images is not known.

### L.2. Independent Pixel Reflection and Detection

All 78,000 originals and their horizontal RGB reflections are processed afresh. Reflection reverses pixel columns at unchanged resolution, with no interpolation, image rotation, finger-index permutation or label substitution. The same MediaPipe model hash and version, IMAGE mode, maximum two hands, and detection/presence thresholds 0.5/0.5 are used in both views. Each image is detected independently. Image landmarks, world landmarks and handedness are taken from its own detector result. This avoids assuming that the detector is exactly reflection-equivariant or that changing a handedness string changes the hand geometry.

Both views use the existing live encoder: 190 normalized image-landmark distances, 630 axis-direction values and 15 world-joint measurements (835 inputs). Exactly one hand must pass the geometric and finite-descriptor checks. In a purely geometric reflection control, image $x$ becomes $1 - x$, world $x$ changes sign, and handedness swaps while anatomical landmark IDs stay fixed. Distances and interior angles are preserved; axis-$x$ direction cosines change sign. Unit tests check that relationship, but the actual experiment reruns detection on reflected pixels rather than substituting ideal reflected features.

Fresh detection accepts 60,251 originals, 60,201 mirrors and 58,941 source pairs with both views accepted. These fresh results have their own provenance; the earlier 97.748% historical-cache score is not the baseline for attributing a mirror effect.

### L.3. Training and Fixed Evaluation

The five historical filename-block partitions are applied to all source IDs, including earlier rejected images. Every derivative inherits its original source group, so no reflected sibling can occur across the training/evaluation boundary. The groups are not verified people or sessions. For each fold, the baseline trains on freshly accepted originals from fitting groups. The candidate adds a successfully detected mirror only for those same fitting sources. Originals whose mirror fails remain; mirror-only recoveries do not enlarge the source population. This separates reflected presentation from adding new source images.

Each source contributes total fitting weight one: 0.5 per view when both are available, otherwise one for its original. Median imputation and angle scaling are fitted on original fitting rows and shared by the two arms. The SVM uses $C = 10$ with its original-fit resolved RBF gamma and balanced class weights; the candidate therefore does not gain a different kernel bandwidth or class prior through duplication. LightGBM retains 100 trees, 31 leaves, learning rate 0.1, minimum child samples 20 and seed 42. Its row-count-dependent tree constraints can still respond to augmentation. No mirror parameters or hyperparameters are tuned on held-out outcomes.

Three presentations use the same held-out sources: originals; their RGB mirrors; and a fixed random mix containing one view per source. Seed 20261106 selects half the sources within each letter/fold for reflection, giving 39,000 originals and 39,000 mirrors. Both models see identical selected images. Selection occurs before model predictions and is not conditioned on detection success. The mix contains 60,225 accepted detections, of which the detector labels 29,584 Left (49.12%) and 30,641 Right (50.88%). Equal reflection frequency is not the same as exactly equal true hand prevalence.

All-attempt results retain failed detections as no prediction (`?`). Macro-precision, recall and F1 average the 26 letter scores,

with undefined scores set to zero. No prediction contributes a false negative for its true letter, so it reduces recall/F1 but adds no false positive. Accepted-only results use the same accepted images for both classifiers. A further paired-both-accepted subset permits comparison of presentations without changing the source population. Source pairs are never treated as independent people or used to claim twice the effective sample size.

*Table 11.* Pooled held-out results over all 78,000 attempted sources per presentation. No-prediction failures are retained. P denotes macro-precision; each source has exactly one selected view in the balanced mix.

| Learner | Presentation | Training | Acc. (%) | P (%) | Macro-F1 |
|---|---|---|---|---|---|
| SVM | Original | Original | 75.505 | 97.539 | 0.84670 |
| SVM | Original | + mirrors | 75.472 | 97.495 | 0.84640 |
| SVM | Mirrored RGB | Original | 17.983 | 26.007 | 0.16554 |
| SVM | Mirrored RGB | + mirrors | 75.226 | 97.293 | 0.84401 |
| SVM | Balanced mix | Original | 46.733 | 73.566 | 0.53849 |
| SVM | Balanced mix | + mirrors | 75.386 | 97.451 | 0.84560 |
| LIGHTGBM | Original | Original | 74.941 | 96.713 | 0.84006 |
| LIGHTGBM | Original | + mirrors | 75.104 | 96.990 | 0.84206 |
| LIGHTGBM | Mirrored RGB | Original | 38.178 | 58.106 | 0.43133 |
| LIGHTGBM | Mirrored RGB | + mirrors | 74.985 | 96.996 | 0.84132 |
| LIGHTGBM | Balanced mix | Original | 56.569 | 78.737 | 0.64104 |
| LIGHTGBM | Balanced mix | + mirrors | 75.051 | 97.012 | 0.84177 |

*Table 12.* Conditional results on successful detection in each presentation. The two training arms use the same accepted evaluation images; the accepted subset can differ between presentations.

| Learner | Presentation | Training | Accepted | P (%) | Macro-F1 |
|---|---|---|---|---|---|
| SVM | Original | Original | 60,251 | 97.539 | 0.97524 |
| SVM | Original | + mirrors | 60,251 | 97.495 | 0.97493 |
| SVM | Mirrored RGB | Original | 60,201 | 26.007 | 0.19249 |
| SVM | Mirrored RGB | + mirrors | 60,201 | 97.293 | 0.97250 |
| SVM | Balanced mix | Original | 60,225 | 73.566 | 0.62650 |
| SVM | Balanced mix | + mirrors | 60,225 | 97.451 | 0.97421 |
| LIGHTGBM | Original | Original | 60,251 | 96.713 | 0.96692 |
| LIGHTGBM | Original | + mirrors | 60,251 | 96.990 | 0.96962 |
| LIGHTGBM | Mirrored RGB | Original | 60,201 | 58.106 | 0.49978 |
| LIGHTGBM | Mirrored RGB | + mirrors | 60,201 | 96.996 | 0.96940 |
| LIGHTGBM | Balanced mix | Original | 60,225 | 78.737 | 0.74139 |
| LIGHTGBM | Balanced mix | + mirrors | 60,225 | 97.012 | 0.96963 |

### L.4. Interpretation and Limits

For SVM, balanced-mix macro-precision changes from 73.566% to 97.451%, and macro-F1 from 0.53849 to 0.84560, with a net correct-count change of +22,349. On originals the F1 comparison is 0.84670 versus 0.84640 (net -26); on mirrors it is 0.16554 versus 0.84401 (net +44,649). The LightGBM balanced-mix changes are +18.275 percentage points in precision and +0.20073 in F1. Read the joint outcomes rather than selecting only the favourable presentation or letters.

The test mix estimates this artificial presentation distribution, not left/right prevalence in actual use. Original/mirror pairs share identity, pose and background. Pixel reflection cannot create independent left-handed signers, natural variations of signing, new viewpoints or new illumination. The detector's handedness labels are not ground truth. No separate left-handed cohort or independent final classification test is available, and this family of filename-block folds has informed earlier development. Scores and fold differences are descriptive, not confirmatory population-level inference. Results of this augmentation do not establish that the explicit hand features themselves improve recognition.

### L.5. Reproducibility and Live Comparison

The supplement `results/mirror_01/` stores the extraction/training protocols, source test mix, all fold and per-letter metrics, detection counts, paired changes, confusion matrices with a no-prediction column, and every held-out prediction. Local checkpoints retain pixels' checksums, source groups, both landmark sets, detector labels and features.

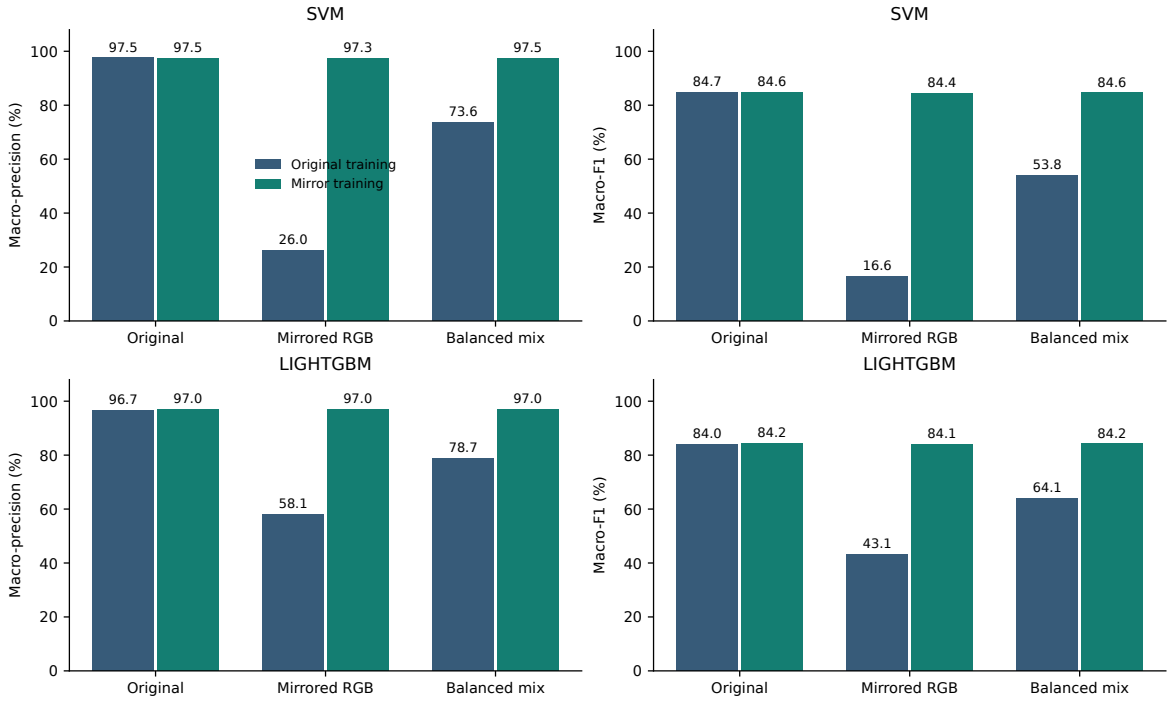



*Figure 8.* Precision and F1 for the two training arms on the three fixed presentations. All attempted sources are retained, including detection failures.

`verify_mirror.py` recomputes metrics and checks source separation, fitting-only preprocessing and model replay. `report_mirror.py` generates paper numbers and tables from stored results. Commands are in `REPRODUCE.md`. These checks establish computational consistency, not independent linguistic validation.

The native live tool exposes a fresh original-only SVM and a mirror-trained SVM as separate comparison models. Both use the unchanged 835-input live encoder. Their deployment fits use all accepted original sources; their cross-validation scores describe the recipe, not an independent test of those final fitted artifacts. The earlier models remain available. A new prospective collection with both hands, hidden predictions, fixed tasks and participant/session separation is needed to test real left-hand performance.

## M. Standalone G/H Vertical-Axis Rotation

### M.1. Question and Separation from Other Experiments

Does adding a $180^\circ$ camera-$y$ turn to G/H fitting examples improve the 26-letter SVM? Neither arm uses mirror augmentation, camera calibration or a routing rule. This differs from RGB mirroring (Appendix L) and the earlier multi-angle recipe (Appendix J). Requested after earlier development, it is exploratory, not an untouched confirmatory study or an angle search.

Observed palm-facing dataset examples and back-facing webcam poses motivated the turn; these views are not verified across all source images. The transformation uses the normalized image-landmark geometry of Shin's directions:

$$R_y(180^\circ) = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \qquad d'_{ij} = R_y(180^\circ) d_{ij}.$$

This equals a wrist-centred landmark turn, $p'_i = p_0 + R_y(180^\circ)(p_i - p_0)$, since translation cancels in pair directions. Unlike horizontal reflection, both $x$ and estimated $z$ change; the determinant is $+1$. Landmark identities, letter labels, 190 normalized distances and 15 intrinsic world-joint measurements stay fixed. No new RGB view or hidden fingers are reconstructed.

### M.2. Matched Fitting and Evaluation

Both arms use the historical 60,251 accepted originals, including 2,430 G and 2,371 H. A freshly fitted reference exactly reproduces all historical held-out predictions. Each G/H fitting source retains its original and adds one turn, weighted 0.5 each; other sources retain weight one. The full fit has 65,052 rows from 60,251 sources. Both are ordinary 835-input, 26-class RBF SVMs with $C = 10$, sharing original-fit imputation/scaling, resolved gamma and balanced class priors.

The same five filename-block folds are retained; original/turned siblings never cross partitions. Each source receives one held-out prediction per arm on its unchanged original. Blocks are not verified people. Scores condition on successful historical detection (77.24% of 78,000 sources), not an end-to-end or independent back-view test.

*Table 13.* Standalone rotation: pooled five-fold scores on the same 60,251 original accepted images. P denotes macro-precision over all 26 classes. Neither arm uses mirror augmentation, camera calibration or a routing gate.

| Training | Accuracy (%) | P (%) | Macro-F1 | Correct |
|---|---|---|---|---|
| Original only | 97.74776 | 97.53902 | 0.97524391 | 58,894 |
| Original + G/H $y180$ | 97.74610 | 97.53874 | 0.97523305 | 58,893 |

The candidate corrects 4 errors and loses 5 correct predictions, net -1: no overall benefit. One fold gains, one is unchanged in correct count and three lose one each. This descriptive comparison on reused folds is not a significance claim. G/H precision includes false positives from all 26 letters.

*Table 14.* Fold-level changes on unmodified original validation images.

| Fold | Reference F1 | G/H $y180$ F1 | Fixed | Lost | Net |
|---|---|---|---|---|---|
| 1 | 0.98137147 | 0.98149750 | 3 | 1 | +2 |
| 2 | 0.93792419 | 0.93793164 | 1 | 1 | +0 |
| 3 | 0.99164126 | 0.99153971 | 0 | 1 | -1 |
| 4 | 0.97810044 | 0.97800902 | 0 | 1 | -1 |
| 5 | 0.97879909 | 0.97874404 | 0 | 1 | -1 |

### M.3. Letter-Level Effects and Known Camera Cases

*Table 15.* All letter-level scores on unchanged original images. "Ref." denotes original-only training; "rot." denotes additional G/H $y180$ turns. G/H recall is unchanged; their precision and F1 decline slightly.

| Letter | Images | P ref. | P rot. | R ref. | R rot. | F1 ref. | F1 rot. |
|---|---|---|---|---|---|---|---|
| A | 2188 | 0.98004 | 0.98004 | 0.98720 | 0.98720 | 0.98361 | 0.98361 |
| B | 2204 | 0.99143 | 0.99143 | 0.99728 | 0.99728 | 0.99435 | 0.99435 |
| C | 1912 | 0.99066 | 0.99066 | 0.99843 | 0.99843 | 0.99453 | 0.99453 |
| D | 2456 | 0.98806 | 0.98805 | 0.97679 | 0.97638 | 0.98239 | 0.98218 |
| E | 2308 | 0.98579 | 0.98579 | 0.99220 | 0.99220 | 0.98899 | 0.98899 |
| F | 2859 | 0.99260 | 0.99225 | 0.98566 | 0.98566 | 0.98912 | 0.98895 |
| G | 2430 | 0.98935 | 0.98774 | 0.99424 | 0.99424 | 0.99179 | 0.99098 |
| H | 2371 | 0.98644 | 0.98602 | 0.98186 | 0.98186 | 0.98415 | 0.98394 |
| I | 2380 | 0.97461 | 0.97337 | 0.96765 | 0.96765 | 0.97112 | 0.97050 |
| J | 2576 | 0.98188 | 0.98188 | 0.96778 | 0.96778 | 0.97478 | 0.97478 |
| K | 2701 | 0.99179 | 0.99179 | 0.98371 | 0.98371 | 0.98773 | 0.98773 |
| L | 2516 | 0.99801 | 0.99880 | 0.99603 | 0.99603 | 0.99702 | 0.99741 |
| M | 1602 | 0.89111 | 0.89158 | 0.91948 | 0.91885 | 0.90507 | 0.90501 |
| N | 1280 | 0.91322 | 0.91396 | 0.87969 | 0.87969 | 0.89614 | 0.89650 |
| O | 2264 | 0.96726 | 0.96725 | 0.97880 | 0.97836 | 0.97300 | 0.97277 |
| P | 2041 | 0.97817 | 0.97817 | 0.98775 | 0.98775 | 0.98294 | 0.98294 |
| Q | 2106 | 0.98942 | 0.98896 | 0.97721 | 0.97816 | 0.98328 | 0.98353 |
| R | 2543 | 0.96874 | 0.96874 | 0.93826 | 0.93826 | 0.95326 | 0.95326 |
| S | 2559 | 0.97145 | 0.97183 | 0.98398 | 0.98398 | 0.97767 | 0.97786 |
| T | 2354 | 0.99060 | 0.99060 | 0.98471 | 0.98471 | 0.98764 | 0.98764 |
| U | 2513 | 0.92106 | 0.92106 | 0.96578 | 0.96578 | 0.94289 | 0.94289 |
| V | 2552 | 0.98317 | 0.98317 | 0.98433 | 0.98433 | 0.98375 | 0.98375 |
| W | 2456 | 0.98211 | 0.98331 | 0.98331 | 0.98331 | 0.98271 | 0.98331 |
| X | 2157 | 0.96859 | 0.96904 | 0.97218 | 0.97218 | 0.97038 | 0.97061 |
| Y | 2580 | 0.99496 | 0.99496 | 0.99380 | 0.99380 | 0.99438 | 0.99438 |
| Z | 2343 | 0.98963 | 0.98963 | 0.97781 | 0.97781 | 0.98368 | 0.98368 |

No saved camera example enters fitting. These previously inspected cases are diagnostic controls, not an independent test; view labels are qualitative. The back-view H is recovered but both back-view G examples remain wrong. All 22 saved cases, including other letters, are retained in the supplement.

*Table 16.* Known selected G/H camera cases. Recognition here does not estimate unseen-view accuracy.

| Observed view | Intended | Reference | G/H $y180$ |
|---|---|---|---|
| Back-view development | G | C | H |
| Back-view development | H | C | H |
| Back-view development | G | A | H |
| Palm control | H | G | G |

### M.4. Limits and Reproducibility

This isolates G/H turns on the original-data recognizer, not interactions with mirroring, light or calibration. Rotating normalized descriptors cannot simulate real visibility, perspective or landmark errors. Source-view scores and selected error recovery do not establish back-view robustness. Feature design and augmentation remain distinct research questions.

`results/rotation_GH_y180_01/` stores the frozen protocol, hashes, all predictions, fold/letter scores, paired changes, confusions and camera replay. Verification recomputes metrics, checks disjoint groups, original-only preprocessing and source weights, replays fold models and reconciles the historical reference. `report_gh_rotation_standalone.py` generates the paper numbers; `REPRODUCE.md` gives commands. These checks support computational consistency, not independent linguistic or anatomical validation.